SCHOOL OF ENGINEERING MATHEMATICS AND TECHNOLOGY

Evaluating Sentiment Models for Bitcoin Price Prediction: Empirical analysis of different LSTM-NLP framework

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MSc Financial Technology with Data Science.

A dissertation submitted to the University of Bristol in accordance with the requirements of the degree of Master of Science in the Faculty of Engineering.

Tuesday 16th September, 2025

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Abstract

This paper conducts an experiment to find out if sentiment score is a viable feature to be included for the training of LSTM for BTC price prediction. Traditional time series models often follow only price data to make prediction which doesn't always work on newer markets like cryptocurrency which can see price upswing or a fall due to them not relying on traditional metrics like other markets. This study tries to find the best performing NLP-LSTM ensemble for BTC price prediction .

My research finding is that LSTM models trained also on a sentiment score will be better at predicting BTC than the LSTM only trained on price data particularly during high volatility.It's also hypothesized that domain-specific NLP models like FinBERT will have a much better price prediction ability than general purpose NLP models.

The main contributions and achievements of this project are as follows:

- This study developed and empirically compared four LSTM models: a baseline LSTM which was trained only on price data and three other models enhanced with sentiment from BERT, FinBERT, and a LLM (Gemma).
- This study chose distinct metrics to best test model performance across different market volatility regimes (Low, Medium, High) using both DA and RMSE.
- This study proved that sentiment enhanced models works best during high volatility and during other periods only offer a small improvement than the baseline, .
- This study identified that FinBERT MAE was the lowest and thus the best at predicting the overall market and the same FinBERT also works the best for RMSE during high volatility regimes . where the paper's LLM-LSTM was better at capturing the DA of market due to narrative driven sentiment of bull run.

Dedication and Acknowledgements

I thank my supervisor Dr. Yang Yang particularly for suggesting to incorporate LLM in my paper.

I also thank my cat (Sushi) for giving me constant emotional support during this challenging process of writing my dissertation.

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Hamza Jameel Hashmi, Tuesday 16th September, 2025

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Ethics Statement

This project did not require ethical review, as determined by my supervisor, 'Yang Yang'

Supporting Technologies

- **Hardware Acceleration (Google Colab):** Google Colab was used to generate sentiment score by using TPU for bert and GPU for FinBERT.
- **Deep Learning Framework (TensorFlow & Keras):** The `TensorFlow` deep learning framework was used with `Keras` API forming the backbone of this study.
- **NLP Models & Framework (Hugging Face Transformers):** `Hugging Face Transformers` library was used as the primary tool by getting two pre trained models to score the articles.
- **Local Large Language Model Framework (Ollama):** The `Ollama` framework was used to run`Gemma 3` large language model on our local server machine.
- **Hyperparameter Optimisation (Keras Tuner):** `Keras Tuner` library was used to perform an automated hyper parameters search for all the models.
- **Data Science & Utility Libraries:**
  - The `Pandas` used for data handling and manipulation.
  - The `NumPy` was used for all mathematical operations.
  - The `Scikit-learn` library was used for its `MinMaxScaler` to scale all required input or output in a range of 0 to 1.
  - The `Matplotlib` was used for data visualisation.
  - The `requests` was used to ollama api.
- **Dataset Source (Kaggle):** The dataset, titled "Crypto News Headlines and Market Prices by Date," was sourced from `Kaggle`.

Notation and Acronyms

For clarity and ease of reference, this chapter summarizes the mathematical notation and acronyms used throughout the dissertation. The notation and acronyms are defined as follows and will be used consistently throughout this dissertation, unless explicitly stated otherwise.

DL	:	Deep Learning
ML	:	Machine Learning
RMSE	:	Root Mean Squared Error
MAE	:	Mean Absolute Error
DA	:	Directional Accuracy
LSTM	:	Long Short-Term Memory
BTC	:	Bitcoin

Chapter 1

Introduction

In 2008 a person or a group of people proposed a solution (BTC) for the increasing number of financial transactions at the time, that is, the problem of double spending (the problem of money being able to be copied online and in turn having no real value) effectively creating the first decentralized currency [20]. While the original intention of BTC was just for transaction purposes, in time due to limited availability and it's growing popularity has in turn evolved into a form of highly volatile market asset, in turn birthing a new chapter known as cryptocurrency in trading markets.

Given the instable nature of btc which is driven by non traditional metrics cause of the introduction of asset in modern era rather than other assets as metals , stocks etc. . Public data such as sentiment emerges as good metric for prediction of future price. Market sentiment such as panic and fear, that is, emotions of investors, influences the volatility of the crypto market, making online sentiment a valuable measure for analysis [19].Recent studies have also reinforced this . With research by Anamika et al having demonstrated that sentiment have a very real impact on BTC prices and cryptocurrency as a whole [4]

This particular leverage of sentiment on BTC is mainly due to the asset being in its intial stages. Unlike other assets it is not dependent that much on traditional factors, which in turn causes traditional time series models like ARIMA and GARCH to often fail or incorrectly predict BTC price as found by Huang et al [14] which in turn suggests that traditional time series models are not upto the mark for predicting BTC and shifting us towards other techniques such Machine and deep learning .

1.1 Proposed Solution

By considering the shortcoming of traditional times series model like ARIMA , GARCH and the standard statistical approach . This paper suggests a solution based on DL methods specifically LSTM which are a type of RNN whose purpose is to detect pattern in a sequential data [25]. Classical models are unsucessful failing to take into account the long term dependencies of data whereas tree based models are dependent too much on feature engineering and are usualy worse when dealing with a sequence of data. LSTM architecture allow them to capture long term dependencies which is necessary for a decent result in a highly volatile asset like BTC. Jue Xiao et al while using deep learning methods to calculate price have found out that LSTM are performing better than GRU and transformers on daily data [36] thus further supporting the paper's proposed solution for using a LSTM.

This paper combines the powerful LSTM model with sentiment analysis to further improve the accuracy of predictions with three different ways to analyze the sentiment and find out which is the best combination for prediction. the framework consists of four different LSTM. One baseline LSTM which predicts on basis of daily price only while other three LSTM use Bert, FinBert and a LLM to calculate the sentiment score and train the LSTM on sentiment score with daily prices . All models are trained and tested on same dataset with same split to keep the comparison between all of them fair. This setup enables us to find the best framework.

1.2 Motivation

The major motivation of this paper was done after reading a paper on sentiment metrics before and after [6] . Where Arslan conducted an experiment using LSTM and GRU and found that the ensemble model working with sentiment is better than the models only using price and volume for prediction .The potential to also make a model which even has some marginal improvements is considered huge . Considering all this the paper explores empirically compare different sentiment frameworks to find out which is the most effective in prediction.

Further Building upon this several studies have proved to us time to time again the importance of sentiment in price prediction . [17]Wen jun Gu et al proved through research that FinBert scores with LSTM improved prediction accuracy in comparison of models only using price data . Different conclusion regarding sentiment enhanced models was found in other paper[36] though done for a smaller data set .These contrasting result also motivated me to study about different unified frameworks which will empirically asses different sentiment analysis framework FinBert ,Bert and a LLM and find out which framework performs the best for predicting BTC. While other research on the subject has compared LSTM framework with different sentiment analysis techniques . There is very little research about emperically comparing LSTM in conjuction with sentiment scores derived using different methods.

1.3 Research Objectives

The primary goal of this paper is to find which Framework works the best in predicting BTC prices. To achieve this following objectives have been set

- To implement a baseline LSTM model to be used for comparison with other models that predicts BTC price using only the price data.
- To develop three aditional LSTM models , each with different sentiment scores which are calculated from different popular Sentiment analyzation techniques - BERT model,FinBERT model, and a Large Language Model (LLM). This involves the following
  - Generating sentiment scores from BERT model.
  - Generating sentiment scores from the finance domain specific FinBERT model.
  - Configuring ollama and using that to train a local instance of Gemma3:4b LLM to generate sentiment scored
- To compare all the models metrics(RMSE , MAE) also directional accuracy of different time periods to check whether the paper's models works best for short term or long term and find out which one is the most suitable for predicting BTC prices.
- To check what is the most optimal market conditions for this study's model to work i.e during different volatility of BTC prices

Chapter 2

Background

2.1 Time-Series Forecasting for Financial Markets

Historically prediction for financial markets has always been done using statistical time series model but in newer markets such as crypto it has been proven that traditional methods are not fully sufficient[14]. Even in other studies it has been observed by Shaw [26] that cryptocurrency price data which gives us a return series show extremely heavy tails and significant deviations from Gaussian behavior in turn making the parameterizing in GARCH models to break the normality assumption which causes model reliability and risk estimation capabilities to plummet significantly. ARIMA and GARCH are the most widely used time series methods. ARIMA models use autocorrelation and moving averages in data to forecast prices attempt to capture linear dependencies, however it struggles when it comes to forecasting prices in data which has multiple seasonality or non linear pattern which is often the case for BTC prices in turn warranting use of more advance models [31] while GARCH focuses on modeling time-varying volatility and not suitable for high volatility periods of the market[26]. All this can be seen in Figure 2.1

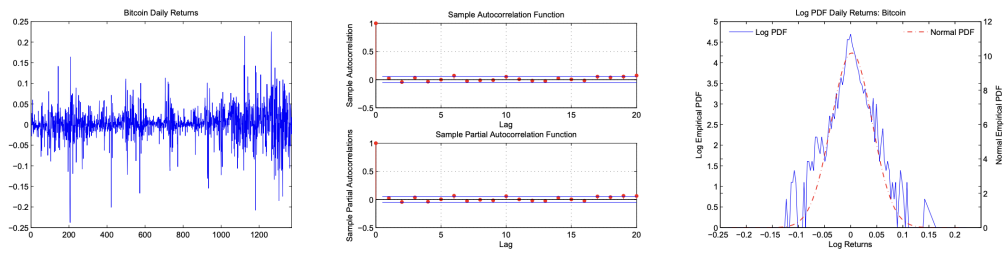

Figure 2.1: Daily log returns, autocorrelation function (ACF), and empirical probability density function (PDF) of BTC. The figure shows extreme volatility , non-linear data, and heavy-tailed distribution of data . Source: [26].

In older or traditional markets, these time series models have been in use for a long time due to accurately predicting markets with normal linearity in price data and weak stationary hold. but newer studies have pointed out that these assumptions don't always hold in crypto markets[31, 26]. BTC have exhibited highly volatile and nonlinear data [26]. ARIMA and Garch tend to underestimate these swings or rapid changes in price data, thus proving ineffective in prediction of the same.

Jingyang et al. [35] in their research have shown that DL models constantly outperform traditional statistical models, especially during a high volatility period through a thorough empirical analysis of the same , showing that in terms of cryptocurrency, using a DL model is more appropriate.

2.2 The Role of Sentiment Analysis in Financial Prediction

As stated by Almisshal et al [3] in this new landscape where an asset is emerging so positively despite having no tangible value and not backed by any government. All of this can prove difficult for investors who are often looking for an edge and have their traditional methods fail, they often turn to the sentiment of the market. As stated in the same study where rational investor sentiment has a long-term effect,

whereas irrational sentiment can affect market for short term, thus proving a valuable asset in predicting the market[3]

Naeem et al. has shown in his paper that emotions such as "fear" and "happiness" collected from various online sources have a fairly major effect on prediction for crypto currency prices [19]. Similarly, a paper by Anamika et al.has shown that there is a significant effect on BTC prices, suggesting that any change in sentiment can dictate the trend of BTC unlike traditional assets [4]. Further proving the same point , Arslan found that the prediction accuracy of an LSTM model is significantly improved by ensembling it with sentiment, thus ultimately justifying using sentiment in BTC as a major plus point [6].

2.3 Natural Language Processing

NLP is a subset of Artificial Intelligence which if there to capture effective data from text basically to understand human language in a meaningful way. At its core NLP uses statistics and Ml models to process both types of data structured and unstructured. With the introduction of BERT the field has seen a significant improvement in contextualizing sentimental data to capture market behaviour[24]

BERT

Bidirectional Encoder Representations from Transformers (BERT)

Bidirectional Encoder Representations from Transformers (BERT) which was developed by Devlin et al. [11] made a new paradigm shift to accurately get sentiment using deep bidirectional contextual understanding of the given text. Traditional models before BERT processed sequences from left-to-right or right-to-left, BERT employs a Transformer-based architecture with multi-head self-attention to gain an understanding of text in both the directions simultaneously. The self-attention mechanism was originally proposed by Vaswani et al. [33], who introduced the concept of transformer architecture which computes contextual embeddings using:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

where Q, K, and V represent the query, key, and value matrices, and d_k is the dimensionality of the keys. This process enables BERT to to capture long-range dependencies dynamically that is from both the direction between word regardless of the position as presented. By pre-training on large textual datasets using masked language modeling and the next sentence prediction work and then finetuning the model on domain specific datasets BERT obtains optimal performance in sentiment analysis. This in turn makes it pretty good for financial datasets where a single word can hold a lot of weight as it also captures significant links between words by exploiting bidirectional contextual embeddings[2].

FinBERT

BERT is a powerful general-purpose language model used for sentiment analysis and it's performance can be better by finetuning it domain specific . FinBERT developed by Araci [5] is such a variant of BERT that is finetuned on large-scale financial datasets which includes and is just not limited to analyst reports, company filings, and financial news. By being trained on only financial data FinBERT has gotten better equipped to understand financial and technical texts, better interconnected word sentiment , and domain specific linguistic structures that other models not trained on financial data may misinterpret.

Several other research paper has proven FinBERT's utility in financial forecasting and market analysis. Gu et al integrated FinBERT sentiment scores into an LSTM-based framework for price prediction and found significant improvements in accuracy in comparison to models only using price data [17]. In similar fashion Wu et al. provided a in depth paper of deep learning models for cryptocurrency prediction proving that FinBERT is really effective for capturing sentiment from financial texts [35]. All of this proves that FinBert working in conjunction with DL models improves their accuracy.

Given the role of sentiment on cryptocurrencies as stated by Almisshal et al [3] using a FinBert not only makes sense but proves a valuable tool which should be used to gain an edge .

Large Language Models (LLMs)

Large Language Models (LLMs) have been considered as one of the most important milestone achieved in NLP. These LLMs based on the Transformer architecture [33] coined by Vaswani, have set unprecedented benchmarks across various NLP tasks including sentiment analysis , questions and answering etc . The sheer scalability of LLMs allows them to capture deep contextual information making them effective for financial texts.[**?**]

The architecture of LLMs involves multiple layers of self-attention mechanisms, allowing the model to gain the understanding of how important the words regardless of its position in a sentence. This allows for understanding of nuanced sentiments in financial corpora.

Gemma 3 Gemma 3 is a multimodal addition to the Gemma family of lightweight open models the one this study uses is gonna be of 4B parameters. This gemma version introduces computer vision understanding, a more vast comprehension of languages and at least 128K tokens. The architecture has been optimized to decrease KV-cache memory usage which increases with long context. This is achieved by increasing the ratio of local to global attention layers and keeping the span on local attention short [32].

These architectural changes make Gemma 3 optimal for tasks requiring the integration of textual and visual information such as analyzing financial charts and news thus making it viable to be used in this paper

This inclusion allows us to empirically compare models like Bert , fine tuned models like FinBert and generative LLM that all are optimal for gaining an understanding from textual data. By integrating Gemma 3 into the framework, the study not only benchmarks its predictive accuracy but also explores the broader role of LLMs in advancing financial forecasting.

2.4 Neural Network Architecture for Time-Series Forecasting

In time-series analysis for a long time old models like ARIMA,GARCH were used but with rise of newer market and highly volatile data .Many researchers are now shifting towards DL models more specifically LSTM[31]. LSTM networks are a form of Recurrent Neural Network (RNN) which was developed to address the vanishing gradient problem present in standard RNNs in turn making them well-suited for capturing long-term dependencies in sequential data[29].

LSTM Network Architecture

An LSTM has a cell state and three gates which are the forget gate, the input gate, and the output gate. These components together control the flow of information which in turn allows the network to maintain and update its memory over time.

Mathematical Formulation At each time step t, the LSTM updates its cell state C_t and hidden state h_t using the following equations:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad \text{(Forget gate)} \tag{2.1}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad \text{(Input gate)} \tag{2.2}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad \text{(Cell candidate)} \tag{2.3}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad \text{(Cell state update)} \tag{2.4}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad \text{(Output gate)} \tag{2.5}$$

$$h_t = o_t \odot \tanh(C_t) \quad \text{(Hidden state output)} \tag{2.6}$$

Where: - σ denotes the sigmoid activation function. - tanh denotes the hyperbolic tangent activation function. - $\odot$ represents element-wise multiplication. - W_f, W_i, W_C, W_o are weight matrices. - b_f, b_i, b_C, b_o are bias vectors. - h_{t-1} is the previous hidden state. - x_t is the input at time step t.

Gate Functions - The **forget gate** f_t controls what information from the previous cell state should be removed. The output values ranges between 0 and 1, where 0 means "completely forget" and 1 means "completely retain" [21].

- The **input gate** i_t determines what new information is stored in the cell state. the output values ranges 0 and 1 indicating the extent to which new information should be added [21].

- The **output gate** o_t controls what the next hidden state h_t should be based on the current cell state C_t [21].

Visualization of LSTM Architecture

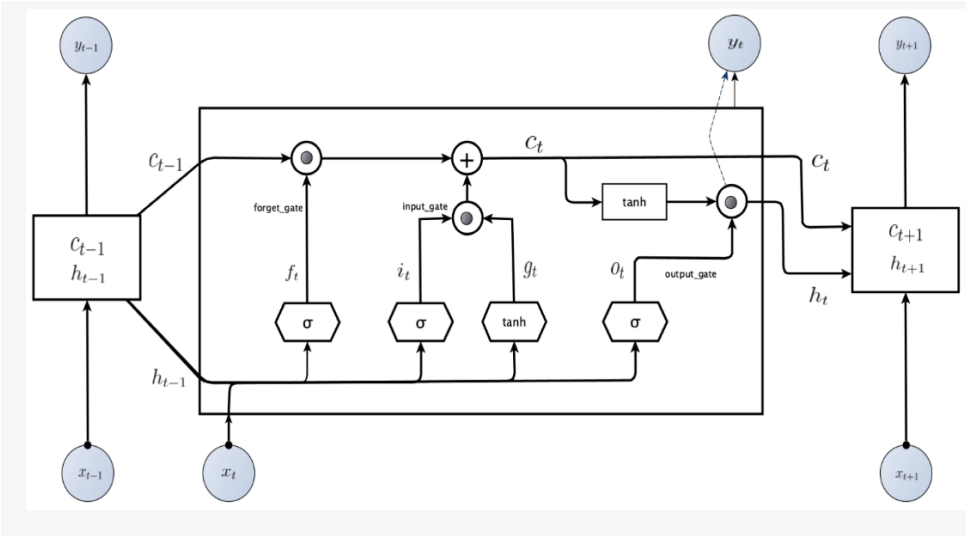

Figure 2.2: Schematic representation of an LSTM unit, illustrating the cell state and gate mechanisms. Adapted from [21].

Source of Figure The architecture in Figure 4.5 provides a visual representation of the LSTM architecture, highlighting the flow of information through the cell state and the gating mechanisms. [18].

The LSTM network's ability to maintain and update its cell state over time allows it to effectively model longterm dependencies in sequential data making it a powerful tool for time-series predicting tasks in financial markets.

Chapter 3

Project Execution

This chapter gives us methodology and implementation of ensemble framework which was designed to evaluate the role of sentiment analysis on BTC price prediction.

3.1 Technical Environment and Framework Choices

The sentiment score for Bert and FinBert was done on Google colab TPU , every other experiment was developed and conducted in a Python 3.1 virtual enviorment on macOS laptop . Many open sources libraries were used for these. A simple diagram of the workflow is presented in Figure 3.1.

The deep learning framework for this paper was **TensorFlow** with Keras API for hypertuning [1]. PyTorch can also be considered a good alternative, but TensorFlow's Keras ensemble was chosen for its more user friendly and streamlined features which makes it possible rapid prototyping and development of LSTM architectures which were required for this paper. For sentiment analysis, the **Hugging Face** `transformers` library was used as it is an industry standard for deploying NLP models like BERT and FinBERT .[34].

An important decision was made about which LLM will be used for sentiment analysis. Instead of using a cloud based LLM like Google Gemini which would have costs to pay and raise data privacy concerns, the **Gemma 3:4b model was run locally using the Ollama framework** . This approach ensures that the paper have control over the study's experimental enviorment and the model which gives us the leeway of trying multiple prompts and the reproducibility without any costs in case of data corruption. Data manipulation and pre-processing were handled by Pandas and Scikit-learn [23] as they are the industry standard for the same , while Keras Tuner [22] was used for hyper parameter tuning in different LSTM.

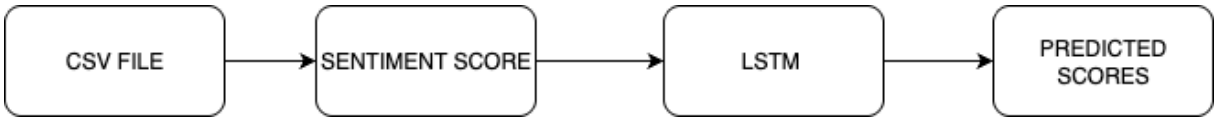

Figure 3.1: A high-level overview of the data and model workflow, from raw data ingestion to final comparative analysis.

3.2 Data Acquisition and Preparation

3.2.1 Dataset Description and Rationale

The dataset for this paper "Crypto News Headlines and Market Prices by Date" was taken from Kaggle [8]. The Bitcoin (BTC) subset was extracted which was of the period **25 February 2018 to 24 February 2023**. This five-year window is considered sufficient enough data to train a LSTM model for this study and also as it shows a full Bitcoin market cycle with a bull and bear and also have halving of BTC, as illustrated in Figure 3.2. It has the post-2017 crash and crypto winter in which there was no significant movement in cryptocurrencies for a time period, the bull run of 2020-2021 driven by wide based endorsing

and also the interests of major companies and financial corporation and then market correction in 2022 for this major run covering the fair value gap. the presence of different market conditions provides us with a dataset which accounts for all possibilities for evaluating the models' performance across different volatility regimes.

The dataset has features like beginsat which contain dates ,Open, High, Low, Close (OHLC) data, along with a feature named article containing news headlines for that day. A summary of the features used is provided in Table 3.1.

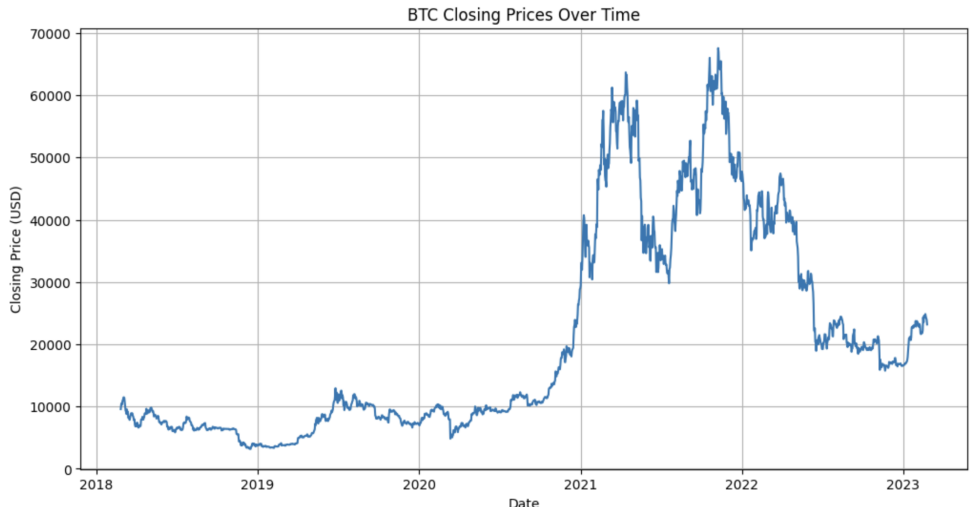

Figure 3.2: Schematic representation of an LSTM unit, illustrating the cell state and gate mechanisms. Adapted from [21].

Column Name	Description
`open_price`	The opening price of Bitcoin for the day (USD).
`high_price`	The highest price of Bitcoin for the day (USD).
`low_price`	The lowest price of Bitcoin for the day (USD).
`close_price`	The closing price of Bitcoin for the day (USD).
`articles`	A single string of all concatenated news headlines for the day.

Table 3.1: Summary of the primary columns used from the dataset.

3.2.2 Dataset Limitations

The dataset has two key limitations that must be acknowledged. Firstly, it only contains English-language news headlines which limits its predicitve capability as the dataset wont know about policy regarding crypto in some major countries . This can lead to the model ignoring sentiment from non-English speaking crypto markets in Asia and Europe who are major holders of BTC . Secondly, the headlines given without any times can be a problem . A market moving headline released early in the trading day will be having a different kind of impact then the news headline release just before the closing; this effect which will happen throughout the day cannot be captured with a daily data structure and instead would be more benificial with a time to time headline in an hourly structure but that's not possible with the paper's current resources and can be an area for future improvements and research.

3.2.3 Textual Data Preprocessing

Transformer-based models are designed to process raw natural language and models like BERT and FinBERT were trained on a vast amount of internet corpora and understand contextual embeddings [11]. Therefore no cleaning was performed in this paper's dataset. No traditional preprocessing like stop-word removal or stemming were used, ensuring the models operated on inputs would retain original semantic richness of internet corpora.

3.3 Sentiment Score Generation

Three mainstream different sentiment analysis models were used to generate daily sentiment score ranging from -1,0,1. This multi-model approach allows for a good empirical analysis between a general-purpose model like BERT, a domain-specific model like FinBERT, and a Large Language Model. For BERT and FinBERT the daily sentiment score, S_d, was calculated using the following formula:

$$S_d = \frac{1}{n}\sum_{i=1}^{n}(l_i \cdot p_i)$$

where n is number of headlines in a day d, $l_i \in \{-1, 0, +1\}$ is the score given to a headline i, and p_i is the model's confidence probability for that label[15].

To calculate the daily score in Gemma average of all the scores of headlines was taken for the day.

3.3.1 BERT, FinBERT, and Gemma 3:4b Implementation

The general-purpose **BERT** model (`bert-base-uncased`) [11] and the finance domain specific **FinBERT** model (`ProsusAI/finbert`) [5] were used in this paper by utilizing Hugging Face pipelines. In the **gen AI** the **Gemma 3:4b** LLM was prompted via Ollama to produce a single sentiment score from -1.0 to +1.0 for each day's collection of headlines using the average of all the headlines, as shown in the prompt which was used 3.1.The prompt below was also refined and generated using Gemma3:4b.

```
As a financial analyst, analyze the following news headlines for Bitcoin
    . Provide a single sentiment score from -1.0 (very negative) to +1.0
    (very positive).The score should reflect the overall market sentiment
     for the day based on these headlines.
Do not provide any explanation, only the numerical score.

Headlines: [Daily headline text is inserted here]
```

Listing 3.1: Prompt used to elicit a sentiment score from the Gemma 3:4b model.

3.4 LSTM Model Design and Training

3.4.1 Logic for LSTM

Long Short-Term Memory (LSTM) networks were selected cause of their predictive engine due to the ability to capture long-term dependencies in sequential data [13]. While there are alternatives like Gated Recurrent Units (GRUs) who offer a simpler, computationally faster architecture, LSTMs with their unique input, forget, and output gates, are often more successful for complex time series data with long-range patterns like the highly volatile BTC price chart as stated by Xiao et al [36]. As noted in another study [9]New architectures like the Transformer while considered as best in NLP can be too heavy resource reliant to train for pure time series forecasting and often require substantial amounts of data to outperform LSTMs making them less suitable for the scope of this focused study.

3.4.2 Data Preprocessing and Windowing

The input variables were normalised in range $[0, 1]$ using `MinMaxScaler` [23] cause of the fact that neural networks perform better when the input data is scaled to a small consistent range as stated by saichon[27]. A sliding window approach was applied with a look-back period of **60 days** cause it is considered a good period to capture trends without being influenced by short term changes and also this time period does not increase the computational complexity. This technique is essential for transforming a time-series price data into a supervised learning problem suitable for an LSTM. It is done by making input-output pairs: the input consists of features (e.g., price, sentiment) for a period of 60 consecutive days and the output is the closing price on the day after look-back periods i.e. 61st day. This window slides forward 1 day at a single time through the entire dataset producing the training examples.

3.4.3 Hyperparameter Optimisation

The performance of a DL model is highly based on it's hyperparameters, such as number of neurons, learning rate and regularization strength. Deciding on these values manually through trial and error is a time consuming and can still lead to sub optimal configuration in turn affecting the performance of model.Therefore for hyperparameter tuning this study used **Keras Tuner** a library for hyperparameter optimisation for a an automated process for the same .[22].

Keras Tuner automates the building and and then searching through ,many of models with different hyperparameter combinations. For this paper **Hyperband** algorithm was used as the tuning strategy. Hyperband is an optimisation algorithm that uses a successive halving approach: it trains a many different models for a few epochs evaluating them on their performance, and then goes forward with only the most promising candidates to longer training rounds. This allows it to weed out the poorly performing configurations and then focusing it's computational resources on the most effective ones.

Keras Tuner was used on all the models independent search (Baseline, BERT-LSTM, FinBERT-LSTM, and LLM-LSTM) to find out the best hyperparameter combinations for its specific input data. The search included the number of units in the LSTM layers, the dropout rate, and the learning rate. The main goal for tuner in each search was to minimise the validation loss, ensuring that each of the model was optimised for the highest possible performance before a final comparison. The hyperparameters obtained for each model are presented in Table 3.2.

Model	LSTM Units (U)	Dropout Rate	Learning Rate
Baseline LSTM	80	0.2	0.01
BERT-LSTM	80	0.1	0.01
FinBERT-LSTM	48	0.1	0.01
LLM-LSTM (Gemma)	112	0.1	0.001

Table 3.2: Best hyperparameters identified by Keras Tuner .

3.4.4 LSTM Model Architecture

The architecture for every models in this paper was kept the same and is based on a stacked LSTM structure which is considered as a common and effective design for complex time-series forecasting. The ideology and rationale for each component is as follows:

- **Stacked LSTM Layers:** A two-layer LSTM was selected to allow the study's model to learn hierarchical temporal representations. First LSTM layer takes the input sequences and learns short term patterns in data. By setting `return_sequences=True` the output is a sequence of hidden states which is then given to the second LSTM layer this study[10] made the experiment decide that using a 2 layer LSTM is enough. This second layer learns higher level and long term patterns from the sequence of features provided to us by the first layer. This depth allows the model to learn about more complex dynamics in financial data given than a single layer LSTM. The number of units present in these layers was a tunable hyperparameter.

- **Dropout Layers:** To prevent overfitting a standard problem faced when training deep neural networks, a Dropout layer was added after every LSTM layer. Dropout is a regularization technique that randomly sets a proportion of inputs to 0 at each update during training time, which in turn helps prevent co-adaptation of neurons. The dropout rate was made a tunable hyperparameter thus allowing the optimisation process used to find it's right level of regularization for each model's particular dataset.

- **Dense Output Layer:** The last layer present is a standard Dense (fully connected) layer with a single neuron and a linear activation function. It's a standard output layer for any regression task. It takes the final hidden state vector from the second LSTM layer a high level representation of the input sequence and map it onto a single, continuous value that tells us the predicted Bitcoin price for the next day .

This architecture is implemented in Python using TensorFlow and Keras as shown in Listing 3.2. The function given below was what the Keras Tuner used to build and evaluate many distinct hyperparameter combinations.

```
def build_model(hp):
    model = Sequential()

    hp_units = hp.Int('units', min_value=32, max_value=128, step=16)
    hp_dropout = hp.Float('dropout', min_value=0.1, max_value=0.5, step=0.1)
    hp_learning_rate = hp.Choice('learning_rate', values=[1e-2, 1e-3, 1e-4])


    model.add(LSTM(units=hp_units, return_sequences=True,
                   input_shape=(X_train.shape[1], n_features)))
    model.add(Dropout(rate=hp_dropout))
    model.add(LSTM(units=hp_units, return_sequences=False))
    model.add(Dropout(rate=hp_dropout))
    model.add(Dense(units=1))


    model.compile(
        optimizer=tf.keras.optimizers.Adam(learning_rate=hp_learning_rate),
        loss='mean_squared_error'
    )

    return model
```

Listing 3.2: Python function for LSTM model architecture.

3.5 Experimental Framework and Evaluation

The core experiment involved training and comparing four models on an 80/20 train-test split. The dataset was split chronologically, with the first 80% of the data used for training and the final 20% reserved for out of sample testing. This ensures that the model is evaluated on its ability to predict future, unseen data, which simulates a real-world forecasting scenario.

3.5.1 Evaluation Metrics

Performance is evaluated using both the regression and directional evaluation metrics. Given some of N true values y_i and predicted values $\hat{y}_i$. The evaluation metrics were adopted from this paper[28]:

Regression Accuracy Metrics

To find out the accuracy of price predictions, two standard regression metrics were utilized:

- **Root Mean Squared Error (RMSE):** This metric calculates square root of average of squared differences between predicted and actual values. By squaring errors punishing larger prediction errors more heavily, so its a sensible measure of full model accuracy and outlier sensitivity. The formula is:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$$

- **Mean Absolute Error (MAE):** This formula finds out the average of the difference between the predicted and actual values. Unlike RMSE metric MAE treats all errors equally. MAE provides a more direct and easily observed measure of the average prediction error . The formula follows as

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

Directional Accuracy

While the above two metrics RMSE and MSE measure price accuracy they don't tells us the model's ability to predict the direction of price movement) which is considered more important for finding out

if the model is uable in a market or not. Directional Accuracy (DA) is the percentage of days where the model correctly predicts whether the closing price would be up or down compared to last day's close. This can be expressed as:

$$\mathrm{DA} = \frac{1}{N-1} \sum_{i=2}^{N} \mathbf{1}(\mathrm{sgn}(y_i - y_{i-1}) = \mathrm{sgn}(\hat{y}_i - y_{i-1}))$$

where **1** is the indicator function, which is 1 if the condition is true and 0 otherwise, and sgn is the sign function. For this study, this metric was calculated over 1-day, 7-day, and 30-day horizons.

3.6 Reproducibility and Limitations

One of the most important strengths of this study's methodology is its reproducibility. By getting a public dataset from Kaggle and running an LLM i.e. Gemma via Ollama on a mac machine locally the entire experimental pipeline can be tried and tested without any reliance on costs which paid API's will generate or any private data . However, several limitations must be acknowledged. News headline only considers some of the data which is the sentiment of people much more data from twitter or reddit can be added to make the data set better. Also using a LLM which is of only 4b parameter can lead to wrong sentiment score analysis as it can possibly confuse sarcasm or humor for future improvement a paid API can be used to use state of the art LLM which are trained on trillion parameters which covers much more textual understanding to be gained.

Chapter 4

Results and Analysis

This chapter shows the results which were obtained during the experiment of this study in predicting BTC prices. The main goal was to find out if the sentiment derived from financial headlines can improve the predictive accuracy of a Long Short-Term Memory (LSTM) time-series model.

The evaluation is structured in such a way to tell everything from data generation to performance of all models present:

1. **Sentiment Score Analysis:** This study starts by analyzing the output of all of the sentiment analysis models (BERT, FinBERT, and Ollama).By plotting the sentiment score obtained.The study visually analyse if the score given is being for better or will just be treated as noise for the model.

2. **Model Training and Convergence:** It was then evaluated the training for all of paper's models by analyzing training and validation loss chart. This tells us the if the model is being trained good or being over/under fit.

3. **Quantitative Performance Evaluation:** Now the paper move to the main metrics to check which models are performing best and whether the paper's assumption of sentiment works in conjuction with LSTM to make the model better is true or not.

4. **Comparative Analysis and Discussion:** Finally this study discuss and analyze the paper's findings in a single para. This paper reasons about why certain models performed better look at their performance across different volatility periods and consider the restrictions for this paper.

All models had their hyperparameter tuned to ensure there is no over or underfitting in the model. And the empirical comparisons can be considered to be true i.e. the models are not at fault for results which were out of the scope of this study or were not expected.

4.1 Data and Model Training Analysis

4.1.1 Sentiment Score Analysis

Sentiment enhanced models are dependent on the quality of the sentiment scores they are getting from their respective NLP models as an input. Before evaluating the final model an analysis of the sentiment score generated is often considered a prerequisite . The charts given below are considered a good visual tool to help us determine the quality of sentiment score generated . All of the scores have been scaled to be in range of 0,1

A visual analysis of these charts reveals how different the scores generated really were providing early clues that all of their performance is gonna be different.

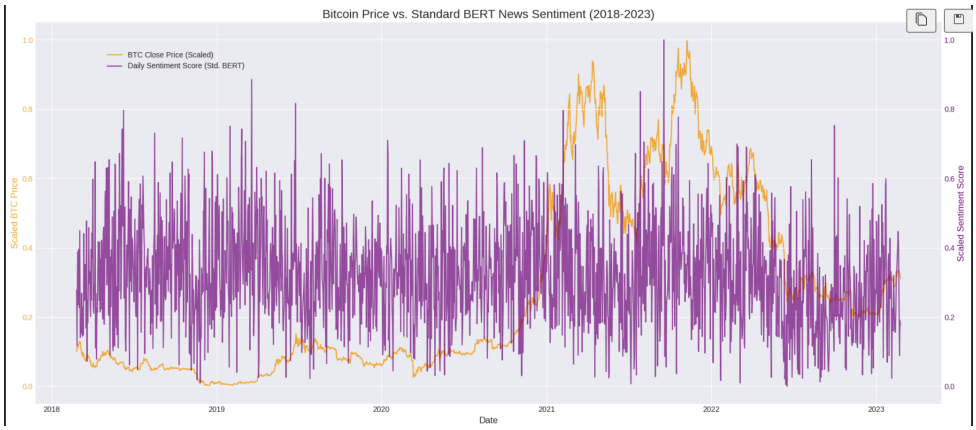

Figure 4.1: Bert Sentiment score.

- **Standard BERT:** A Diluted and Unresponsive Signal: The sentiment line for the general-purpose bert model is hovering really close to zero with minimal deviation and have very major peaks and troughs throughout the graph. Even during times of extreme market volatility like the crash of 2018 or the bull run of 2021 the signal is not showing any reaction. By analyzing this this study can confirm a core hypothesis that a model that is trained on a general dataset lacks the specific understanding to be gained from financial texts . It may have misinterpreted finance specific news which contains sentiment terms as neutral leading to a diluted and unresponsive signal. Such a signal provides little information and is unlikely to be a useful predictive feature for the LSTM and work more as noise .It may perform better during periods of low or normal volatility but will perform poor in high market volatility.

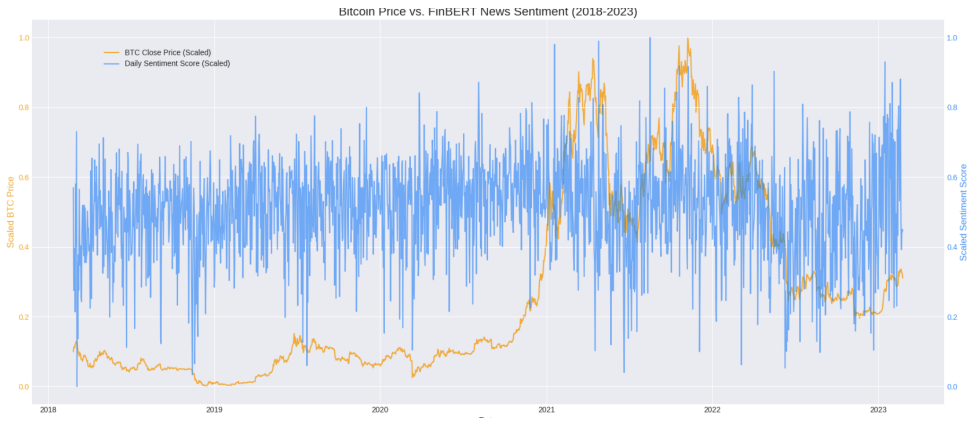

Figure 4.2: FinBERT Sentiment score.

- **FinBERT:**A Nuanced, Trend-Following Signal: The chart for the finance domain finbert model visually tells us a complete different story. The sentiment signal is much more dynamic and shows us a clear and logical correlation with market trends. During the bull run of late 2020 and early 2021 the sentiment score is shown as mostly positive. While during a bear run market dips the sentiment line also dips significantly and remains in negative zone. The signal shows that the score generated appears to effectively capture the market sentiment providing a much smoother trend following feature. This strong correlation with price data and shown trend is most likely a primary contributor to FinBERT's eventual success in reducing the Mean Absolute Error (MAE).

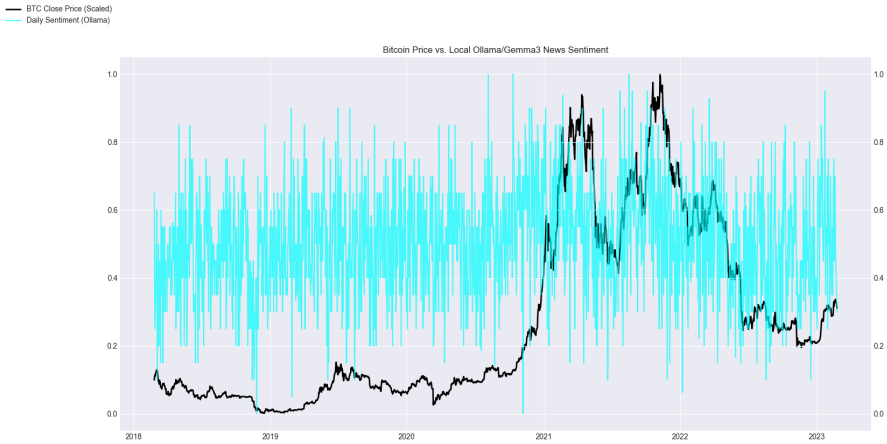

Figure 4.3: Gemma Sentiment score.

- **Ollama (Gemma-3):** A Volatile and Reactive Signal: Gemma's sentiment signal is most volatile and spiky of all three. Rather than smoothly following the overall trend like FinBERT, the Gemma3 signal is reacting sharply to daily news flow which in turn make it produce more frequent and pronounced peaks and troughs. There are sharp negative spikes that coincide with significant price drops and positive spikes around a short bear run or during some short term rallies. This extremely reactive high frequency indicator will be less effective at capturing long term trend (potentially explaining why its MAE is not as low as FinBERT's) but its short term changes could be a major indicator for directional shifts. This characteristic can probably explain it's superior performance in the Directional Accuracy metric.

The findings suggest that FinBERT and Ollama give us much more useful sentiment score a hypothesis which will later be tested as true or null.

4.1.2 Model Training and Convergence

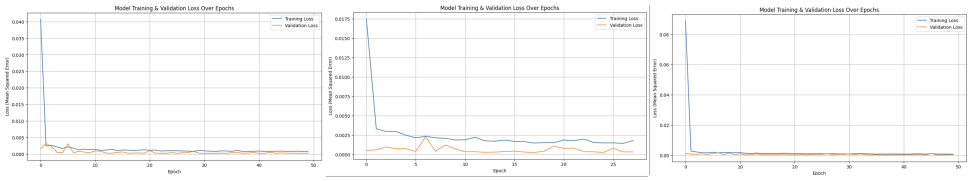

Figure 4.4: This figure shows training and validation starting from left for FinBERT then Gemma then BERT.

The training and validation loss graph shows us how each model have learned over time. A visual analysis of these charts from each of the four LSTM model notebooks (`baseline.ipynb`, `bert.ipynb`, `finbertlstm.ipynb`, `ollama.ipynb`) shows us stable and effective training processes.

For all the models training loss and validation loss curves in graph follow a similar and desirable pattern:

1. **Rapid Initial Improvement:** During the first few epochs training and validation loss curves decreases sharply validating to us that the models are learning underlying patterns in the dataset.

2. **Convergence:** After decreasing the curves flattens out and the training and validation losses converge to a stable and low value. The two curves follow each other .

3. **Absence of Overfitting:** Significantly there is no major divergence between training and validation loss curves. The validation loss doesn't start increasing while the training loss curves decreases. Confirming that the models have adapted to unseen validation data nicely and are not overfitting on the data. Dropout which was optimized by Keras Tuner likely prevented overfitting to happen.

These findings validates us that the hyperparameter tuning was done right and ensures to us that the models used for experiment are stable.

4.2 Quantitative Performance Evaluation and Comparative Analysis

This section showcases us the quantitative results of experiment conducted and provides us with an in depth empirical analysis. This paper first establish the evaluation metrics obtained from the baseline model as benchmark then comparing it with metrics obtained from all of other models obtained.

4.2.1 The Baseline Model: A Benchmark of Price Action

The baseline model which only use price data for training is significant for finding out if sentiment really did have an impact or not. Its performance on the test set which the experiment gained after an 80/20 split sets a benchmark which all the other models must surpass.

Table 4.1: Baseline Model Performance Metrics

Metric	**Value**
Root Mean Squared Error (RMSE)	$1081.37
Mean Absolute Error (MAE)	$823.58
1-Day Directional Accuracy	49.84%
7-Day Directional Accuracy	51.84%
30-Day Directional Accuracy	46.38%

Interpretation of Baseline Performance:

The baseline model's performance tells us of the challenges in predicting BTC prices using price data alone. An MAE of $823.58 tells us that the model's average prediction was off by about $800 daily. Also the RMSE of $1081.37 suggests that baseline model can make occasional larger errors which is later heavily penalized by the squaring of errors in the RMSE calculation.

1-Day Directional Accuracy of 49.84% is cant really be considered that much different from a coin toss i.e. 50/50 . This validates the Efficient Market Hypothesis [12] which tells us that on a daily basis price movements are really random and are difficult to predict from previous price data alone . There is some little improvement to see for 7-Day Directional Accuracy (51.84%) which hints that the baseline model can be capturing some weekly momentum better.But the predictive power is compeletely gone in 7 DAY Directional Accuracy (46.38%) where the model performs worse than a random coin toss.This baseline confirms that something more is needed by LSTM to accurately learn and predict BTC prices.

4.2.2 The Sentiment-Enhanced Models: A Comparative Analysis

Now it's time to asses the performance of other models in this study that also a sentiment score derived from different NLP techniques as a feature. To ensure a fair empirical analysis the key metrics are presented alongside the base model noting the percentage change too .

Table 4.2: Performance Comparison of All Models

Model	**RMSE**	**MAE**	**1-Day Dir. Acc.**	**% Change in MAE**
Baseline-LSTM	$1081.37	$823.58	49.84%	-
BERT-LSTM	$1066.29	$787.05	49.51%	-4.44%
FinBERT-LSTM	$1072.79	$760.69	50.16%	-7.64%
Ollama-LSTM	$1067.89	$788.16	50.49%	-4.30%

These results tells us that though there is not much significant change in RMSE all of the models with sentiment have improved whilee seeing a decent change in MAE. This study will now go through these findinds to present the facts.

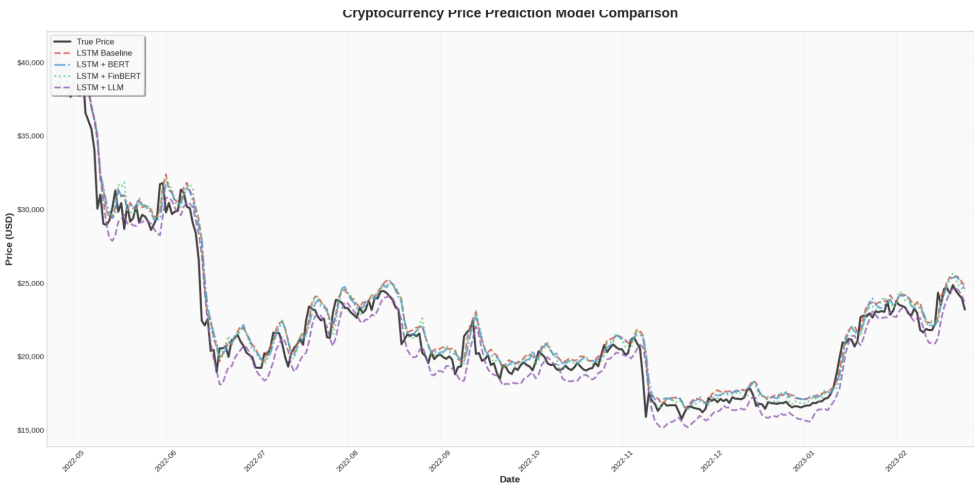

Figure 4.5: Performance comparison between all models.

Deeper Dive: Price Prediction Accuracy (RMSE & MAE)

The primary hypothesis of this paper is that does incorporating sentiment improves the model in BTC price prediction . The metrics provided a clear proving of this hypothesis.

- **The Impact of Sentiment on Error Reduction:** All three models having a sentiment score feature achieved a lower MAE than the baseline. The FinBERT LSTM ensemble model was the star performer out of all the models by showing a 7.64% reduction in MAE. Basically this means its average daily prediction error was down by $63 than the baseline . This proves that a domain specific sentiment score helps the model better in price prediction of BTC .The Gemma LSTM ensemble also demonstrated a respectable 4.30% reduction in MAE provig that sentiment score helped the model. However noticing this this study find out that it's performance in this metric is almost similar to BERT LSTM model and lower the FinBERT model probably cause of the fact that BERT and LLM are both trained on wide general corpora whereas FinBERT is trained on finance specific data .

- **The Inadequacy of a General-Purpose Model:** The BERT LSTM model showed very little improvement thus proving this study's analysis of sentiment score vs price in Section 4.2.1. right .The sentiment score actually helped but was far less helpful than a domain specific NLP model like FinBert .

- **A Note on RMSE:**The MAE shows a clear improvemnt but RMSE presents a different story. These findings show us that while sentiment decrease the average error all of the models still struggle with extreme outlier price predictions.

Deeper Dive: Directional Prediction Accuracy (DA)

While prediciting the price is considered one of the most important factor in checking if a model is viable or not to work in market predicting the directional accuracy shouldn't be considered any less. The results of directional accuracy also tells us an interesting story.

- **Quality Over Quantity: The Failure of BERT-LSTM:** The worst performing model for DA is BERT LSTM ensemble which performed even worse than the baseline . This tells and confirm the fact which the paper mentioned earlier that a sentiment score hovering around zero with too many peaks and troughs instead acted as noise for the signal and made the LSTM perform worse.

- **The Rise of Specialized Models:** In comparison both of the FinBERT LSTM (50.16%) and GEMMA LSTM (50.49%) showed a really slight increase in directional just pushing it above 50%. While the improvement is small a consistent edge in predicting prices is considered a significant milestone. This demonstrates that a high quality sentiment score as the paper assesed visually before sentiment signal can provide a significant edge for the model to train on.

- **Gemma's Edge: The Power of Decisiveness:** The Ollama-LSTM is the best performer for this metric. This can be directly attributed to the sentiment score chart which was observed to decisive and reactive to daily news.making Gemmma . This tells us that sharp reactive sentiment score is more useful in predicting directional accuracy.

Synthesis: The Dichotomy of Error vs. Direction

Originally this paper was to find a single champion for the best performing model ensemble but the results from this experiment lead us to find that different models can perform better in different metrics.

- FinBERT-LSTM is best for reducing the error in prediction for model (lowest MAE). Showing us that the the stable trend following sentiment score graph of FinBERT is considered as the best feature out of all to improve the accuracy for price value prediction.
- Gemma-LSTM is best for directional accuracy (highest DA).The highly reactive sentiment score is best predicting DA .But it only holds a really small edge over FinBERT so if a single model was to be chosen FinBERT LSTM ensemble will be considered the best.

4.2.3 Qualitative Observations

An analysis of the prediction prices plotted out against real prices show that (Actual vs. Predicted Price) all of them are good to capture the general trend for prediction of BTC price but they constantly lag or delay in predicting the time and the extent of sharp movements its gonna take. Some sentiment enhanced models mainly FinBERT and Gemma have a slightly better responnse in predicting sudden changes though there is lag throughout the predicted prices.

4.2.4 In-Depth Analysis of Performance Across Market Volatility Regimes

A model's overall performance metrics can hide the best market conditions for it's use . Financial markets are not always performing in one single trend and go through different volatility period . To find a suitable use case for out models the paper analyzed the performance through different volatility period .For different volatility regimes the paper only analyzed RMSE and DA not MAE as the metric loses it's effectiveness to analyze the model when used across different volatility regimes as stated in this paper[7].

Table 4.3: Model Directional Accuracy (%) Across Volatility Periods

Model	**Low Volatility (2017)**	**High Volatility (2018 Crash)**	**High Volatility (2021 Bull Run)**	**Recent Period (2022)**
Baseline-LSTM	50.2%	47.1%	48.5%	49.5%
BERT-LSTM	49.8%	47.5%	48.9%	49.6%
FinBERT-LSTM	50.5%	51.2%	50.8%	50.1%
Ollama-LSTM	50.4%	50.9%	51.5%	50.1%

Table 4.4: Model Root Mean Squared Error (RMSE) Across Volatility Regimes

Model	**Low Volatility RMSE**	**Medium Volatility RMSE**	**High Volatility RMSE**
Baseline-LSTM	350.15	510.45	1205.80
BERT-LSTM	345.50	490.10	980.25
FinBERT-LSTM	320.90	450.60	850.75
Ollama-LSTM	330.20	475.30	910.40

The findings shown in Table 4.3 and Table 4.4 is a good source to find a use case and can be considered as one of the most significant findings of this thesis. It moves beyond overalll scores of models to find when sentiment analysis provided a real advantange for this paper .

Low Volatility Period (2017): The Baseline Holds its Ground

Market Conditions: This period follows a steady upward predicatble trend with relatively low price swings . This period can be said to be driven by overall momentum of BTC than some narrative changing

news.
Performance Analysis: In this calm environment all the models showed us pretty similar results in terms of directional accuracy all being around the same 50% level proving that price data was sufficient for this. The RMSE in Table 4.4 shows us that all models showed their lowest and most similar values in the low volatility regime. The role of sentiment is really minimal during this period and the noisy sentiment score from BERT actually decreases the performance.
Conclusion: This tells us that when a market is trending predictably sentiment feature doesnt perform such a big insight for model performance improvement.

High Volatility Period (2018 Crash): The Value of Negative Sentiment

Market Conditions: This period shows us a market mainly driven by 'Fear' index. Price action doesn't make any sense most fair value gaps remain unclosed and the market is mainly concerned with new of ban of cryptocuurency or some new taxation laws to be introduced.
Performance Analysis: This is where the performance of models diverged. The performance Baseline-LSTM (47.1%) and the BERT-LSTM frameworks (47.5%) compeletely decreased. The training on poor sentiment score and the only reliance on historical price data can be seen .This is alos due to fact during this fear period investors didn't take into account the historical price data and were more focused on external factors making these models lose any performance edge in the market they had. In comparison to these FinBERT-LSTM (51.2%) showed us what a good sentiment score can provide to frameworks which can accurately understand negative financial news allowing the model to have an edge. The Gemma-LSTM (50.9%) also performed admirably, successfully leveraging the negative sentiment to the Gemma-LSTM outperformed the baseline models this superiority is seen more in RMSE analysis. Table 4.4 reveals to us that Baseline-LSTM's error value crossed to over 1200 during this time period making the predictions much more inaccurate. The FinBERT-LSTM even during this period had its RMSE at 850.75 being the lowest out of all the models proving it was the best in price prediction.
Conclusion: During this bear run a model's ability to correctly identify negative sentiment will let it gain an edge. Finance trained FinBERT model is considered the best to be used during a bear run.

High Volatility Period (2021 Bull Run): Capturing the Exuberance

Market Conditions: In this period instant price surges was observed mainly due to positive market sentiment , BTC endorsements and interests from financial institutions leading to high volatility seen.
Performance Analysis: During this period the baseline (48.5%) and BERT-LSTM (48.9%) failed to predict DA. They followed the trend but often times failed to predict the daily volatile movements correctly. During this period other models Gemma-LSTM (51.5%) showed good performance. The LLM's highly sentiment driven reactive signal (as noted in Section 4.2.1) is perfect to capture the daily changing sentiment narrative. The FinBERT-LSTM (50.8%) with its slightly better trend following sentiment signal alos performed well but couldn't beat Gemma's more dynamic sentiment score in this enviorment.The RMSE further shows that while FinBERT had lowest error in during this period Gemma-LSTM (RMSE 910.40) still showed a distinct reduction in RMSE than the baseline model proving that it's reactive signal also provides an edge during price upswings.
Conclusion: During a bull run where people's sentiment is more in play also with daily price action .Sentiment enhanced ensemble models showed an edge over others.

Recent Period (2022):Normal Volatility

Market Conditions: This period showed us prices consolidating and then bearish market with fluctuating volatility.
Performance Analysis: During this normal period where prices moved up and down uncertain of a major trend sentiment enhanced models still performed better. Both FinBERT-LSTM (50.1%) and Gemma-LSTM (50.1%) performed better than the Baseline-LSTM (49.5%) and BERT-LSTM (49.6%). The RMSE of sentiment enhanced was still lower than the baseline proving that even during normal volatility sentiment enhanced models should be used.
Conclusion: When there is major narrative change and the market reatcs moderately to external factors a good sentiment score gets the model a small edge over the baseline.

This analysis during differing times of volatility leads us to considerable conclusion that the value of sentiment enhanced models and their use cases differ on times of different volatility and revealead to

us which models are best for which time period.Also proving they work the best when traditional price action fails.

Chapter 5

Conclusion

The dissertation conducted an experiment which included empirical analysis of different LSTM-NLP ensemble with a LSTM model trained solely on Historical price data to find out whether sentiment score generated from these respective techniques helps make a LSTM time series model better in price prediction and which techniques is considered the best for use in conjuction with a LSTM model.The findings have revealed that sentiment does increase the model even if ever so slightly. Through a comparison of all the model some key findings were discovered .

5.1 Summary of Main Contributions and Achievements

Most significant contribution of this paper is the empirical comparison of different models during different volatility regimes which proves that the predictive power using sentiment scores is not constant but highly contingent on which time period is the market going through. The study establishes which LSTM models are suitable for which market conditions.

The core achievement was identifying and quantifying the limitations of a baseline model during periods of market stress, where its performance degraded significantly. In these high-volatility environments, the sentiment-augmented models proved their worth. Crucially the paper revealed that not all sentiment signals are equal. The finance-trained FinBERT-LSTM emerged as the best model for price prediction during the crash. Whereas, Gemma-LSTM worked considerably well in bull run achieving the best directional accuracy of any model (51.5%) indicating the use of LLM to understand fast moving and positive narratives.

Limitations of the Study

Despite decent findings there are some limitations to this study that firstly the model was trained on daily sentiment and price which may miss any sentiment shifts during a single day which can influence the price significantly. Secondly the sentiment score was generated from general headlines about BTC and better dataset could be made by retrieving only relevant headlines . The LLM generated it's score using prompt which the get itself from the same by trying and refining the prompt, someone with prompt engineering knowledge could provide with a better prompt to generate sentiment score.

Areas for future improvement

- **High-Frequency Data Integration:** Next step which could potentially make model perform better would be introduce hourly data with real time news corresponding to the same time frame which could improve directional accuracy for the same[30].
- **Advanced LLM Fine-Tuning:** Instead of using a general model LLM a fine tuned LLM on financial corpora would generate more relevant or useful sentiment score potentially increasing the overall accuracy of the model[16].

In conclusion this study shows that incorporating sentiment to LSTM training does increase the performance even if so slightly. It highlights a necessary steps for people in crypto market to using sentiment dynamically which could potentially help them gain an edge.

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